

LESSON GUIDE

Technology Plus

Unit 7: Security

6.3.1 – Password Best Practices

Lesson Overview:

In this lesson, students will explore the best practices for having safe and secure passwords.

Students will:

- Discuss the best practices when creating a password.
- Create and test the strength of passwords.

Guiding Question: How can following password best practices keep my accounts safe and secure?

Suggested Grade Levels: 8-12

Technology Needed: Yes

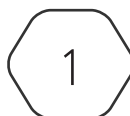
Approximate Time Required: 45-60 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 6.3 – Explain password best practices.
 - Password length
 - Password complexity
 - Password history
 - Password expiration
 - Password reuse across sites
 - Password managers
 - Password privacy
 - Password reset process
 - Changing default usernames and passwords
 - Enabling passwords

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Lesson Background

Background Information:

Passwords are one of the oldest and most common security tools in the digital world. They act as digital keys, protecting everything from social media accounts to banking information, school portals, and even smart devices. Without strong passwords, attackers can easily guess, crack, or steal them, giving them access to sensitive personal information. A strong password is not just about random characters - it's about length, complexity, uniqueness, and privacy. Tools like password managers help people create and remember strong passwords without relying on memory alone. Understanding how to reset and manage passwords properly, and why companies enforce rules like expiration or history checks, helps users stay safe.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 6.3.1 – Password Best Practices • Activity 6.3.1 – Password Power
Materials per Student	• Student Notes 6.3.1 – Password Best Practices • Student Handout 6.3.1 – Password Best Practices • Assessment 6.3.1 – Password Best Practices (optional)

Cyber Terms:

Term	Definition
password	a secret word, phrase, or string of characters that proves your identity and lets you log in to an account, system, or device
passphrase	a longer password made up of multiple words or a full sentence; passphrases are easier to remember but harder for others to guess
password managers	a secure app or tool that creates, stores, and remembers strong, unique passwords for each of your accounts – so you only have to remember one primary password

6.3.1 Password Best Practices

Guiding Question: How can following password best practices keep my accounts safe and secure?

Lesson Launch (5 minutes)

- Imagine someone gained access to your phone, email, social media, and other online accounts. What is the worst thing they could do?
- Give students time to think and allow a few students to respond.

Students should conclude that someone could impersonate them, leak their private information, mess with their assignments, steal their money, etc.

- Passwords are the first line of defense against this. Without a strong password, someone could get into your accounts and obtain your personal private information

Don't Get Cracked! (20-25 minutes)

- Distribute the Student Handout 6.3.1 – Password Best Practices and have students fill in the answers during the lesson.
- Introduce a **password**, its definition, their purpose, and what students will learn in this lesson.
- Discuss password length and complexity and critical factors to creating a safe and secure password.
- Tips for creating a safe and secure password:
 - Avoid Common Words and Patterns: Easy to guess words, phrases, or predictable sequences should be avoided, e.g. 'Password123' or 'QWERTY!@#'.
 - Avoid Using Personal Information: Don't include any obtainable information about you like birthday, names, or common phrases.
 - Use Passphrases: A **passphrase** consists of a sequence of words or a sentence. It is easy for a user to remember but difficult for someone to guess, e.g. P!nk\$L0th\$3@tL0t\$0Ffru!T
 - Randomness: Use a password generator to create a random mix of characters.
 - Multi-Factor Authentication (MFA): MFA adds an additional layer of security by requiring a second form of verification for a user to access their account.
- Tips for managing passwords:
 - Use a Password Manager: Password managers can be used to generate and store passwords, making it easier to have a unique, long password for each account without needing to remember them all.
 - Regular Updates: Passwords should be changed often to reduce compromise in the event of a leak. If a password is compromised, it should be changed immediately.
 - Avoid Reuse: Use a different password for each account.
- Discuss password history.

- This policy is more common in business or enterprise settings than in everyday sites. Encourage students not to reuse old passwords, especially ones that were weak or shared.
- Discuss password expiration.
 - Most everyday sites don't have this policy. It is often found in business settings where confidential information is handled. Students should still change their passwords if they believe an account may be at risk or has been exposed.
- Discuss password reuse across sites.
 - This is a very common practice but should be discouraged. Tell students they can better protect their accounts if they have different passwords for all of their accounts.
- Discuss password managers.
 - Most web browsers and phones have password managers built-in that will ask users if they want to save their password. If a user says yes, the password is typically available for autofill the next time they visit the site.
- Discuss password privacy.
 - Students at this age often share password with close friends or family. Tell them that passwords shouldn't be shared, especially if it is protecting personal and private information.
- Discuss the password reset process.
 - Students may have already followed this process when they forgot a password. Remind them that if their current passwords don't meet best practices, they can use this process to update them now.
- Discuss changing default usernames and passwords.
 - Some devices or accounts come with a default username and password, these should always be updated immediately.
- Discuss enabling passwords.
 - Students likely have devices, apps, or accounts where they have not enabled passwords. Encourage students to check today if there is any place where they should reconsider enabling a password. This can include smart watches, tablets, apps that hold personal information, and more.

Activity: Password Power (20-25 minutes)

- Display Activity Slides 6.3.1 – Password Power, have students record their answers in the handout.
- For this activity, students will use our password strength tester <https://cyber.org/find-curricula/test-strength-your-passwords> in a browser.
- If for some reason students cannot access the password strength tester in their web browser, have them login to the CYBER.ORG Range and use the Ubuntu Machine to access the tester from there.
- Students create sample passwords, test them, and record the strength level and estimated time to crack.
- Remind students: Never test their real passwords in an online password tester.

Activity Debrief Questions and Answers

Sample Passwords	Estimated Crack Time	My Versions	Estimated Crack Time
dog789	51 minutes		
Basketball#1987	121 days		
T@C0Tu3\$d@yB@by	36 centuries		
WorkingITIsTheBest!	4764 centuries		

Activity Debrief Questions and Answers

1. Which types of passwords get cracked instantly?

Passwords that are short with no complexity.

2. Which types of passwords take years or centuries to be cracked?

Passwords that are longer and include complexity like symbols and numbers.

3. What tips can you take from this to improve your real password?

Make your password something memorable to you but hard for others to guess – like a phrase with numbers and symbols sprinkled in.

Putting It All Together (5 minutes)

- Review what students learned today by discussing the questions below as a class.
- What should every password include?

Every password should include uppercase letters, lowercase letters, numbers, special characters or symbols, and be at least 12 characters long.

- What policies keep passwords safe and secure?

Password history prevents reusing old passwords. Password expiration requires users to change passwords after a set time.

- What could happen when users reuse their password for multiple sites?
If one account is breached, attackers can use the same password to access all other accounts.
- When should a user reset their password?
A user should reset their password if it is short, only letters, reused on other accounts, protects sensitive data, was exposed in a breach, or is a default password.